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CLOUD ENGINEER INTERVIEW

9 CLOUD CONCEPTS TO CRACK THE INTERVIEW

For Cloud Engineer interviews: the ideas panels probe, explained the way you should answer them.

1.

Shared Model

The cloud provider secures the platform; you secure what you put on it.

- Provider: hardware, hypervisor, physical network
- You: data, identity, config, app code
- Line shifts left as you move IaaS -> PaaS -> SaaS

In an interview, say: 'Security in the cloud is a partnership, and misconfiguration is usually on our side of the line.'

2.

IaaS/PaaS/SaaS

Three service tiers, three levels of control vs convenience.

- IaaS: VMs, disks, networks – you own the OS
- PaaS: managed runtime – you own the app
- SaaS: finished product – you own the data + users

In an interview, say: 'I pick the highest tier that still meets the control and compliance need.'

3.

Regions & AZs

Regions give you geography; availability zones give you fault isolation inside a region.

- Region = geographic location, latency + residency
- AZ = isolated datacenter within a region
- Multi-AZ for HA, multi-region for DR

In an interview, say: 'Latency, data residency and blast radius drive my region and AZ choices.'

4.

Scale Up vs Out

Vertical adds power to one node; horizontal adds more nodes behind a balancer.

- Up: bigger VM, simple, single point of failure
- Out: more instances, resilient, needs stateless design
- Autoscale on CPU, queue depth or custom metrics

In an interview, say: 'I default to scaling out – it's cheaper, more resilient, and cloud-native.'

5.

LB + Stateless

Load balancers only work when any request can hit any instance.

- L4 for TCP/UDP, L7 for HTTP routing and WAF
- Push state to cache, DB or object storage
- Health probes remove bad nodes automatically

In an interview, say: 'Sticky sessions are a smell - I keep the app stateless and the state in a store.'

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6.

IaC

Infrastructure defined in code, versioned in Git, applied by a pipeline.

- Terraform, Bicep, CloudFormation, Pulumi
- Declarative + idempotent, no click-ops
- Enables review, rollback and repeatable envs

In an interview, say: 'If it isn't in Git, it doesn't exist - that's how I keep envs consistent.'

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7.

VNets & Net

Virtual networks segment workloads and control what can talk to what.

- Subnets, route tables, NSGs / security groups
- Private endpoints keep PaaS off the public net
- Peering + VPN / ExpressRoute for hybrid

In an interview, say: 'I design networks least-privilege first – default deny, open only what's needed.'

8.

Identity + ZT

In the cloud, identity is the new perimeter - assume the network is hostile.

- Managed identities kill secrets in code
- RBAC + least privilege, scoped to resource
- Zero Trust: verify explicitly, assume breach

In an interview, say: 'No stored keys - workloads authenticate as themselves via managed identity.'

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9.

Obs + Cost

You can't operate or optimize what you can't see - logs, metrics, traces and spend.

- Metrics + logs + distributed tracing = the three pillars
- SLOs and alerts on user-visible symptoms
- Tag everything; right-size, autoscale, reserved capacity

In an interview, say: 'Observability and FinOps are the same habit - measure, then decide.'